Supplementary material to

Explainable production-adjusted electrical-energy baselines for industrial feed pelleting

Vladimir Sousa Santos[a], Juan José Cabello Eras[b],
Eder Elías Herrera Rodiño[c], Francisco Manuel Arrabal-Campos[d,*]

[a]Department of Energy, Universidad de la Costa, Barranquilla, Colombia

[b]Department of Mechanical Engineering, Faculty of Engineering, Universidad de Córdoba, Montería, Colombia

[c]Master's Programme in Mechanical Engineering, Universidad de Córdoba, Montería, Colombia

[d]Department of Engineering, Research Centre CIAIMBITAL, Escuela Superior de Ingeniería, University of Almería, Spain

*Corresponding author: fmarrabal@ual.es

S1 Contents

This document complements the main text with material that was not reproduced there: the complete source-variable dictionary and descriptive statistics of the industrial records (Section S2), the encoding and the full coefficient table of the frozen Ridge model (Section S3), the reference estimators and the development comparison at full precision (Section S4), the block-bootstrap and multiplicity procedures (Section S5), the calibration arithmetic of the residual intervals with line- and week-level diagnostics (Section S6), the protocol and stratified results of the matched temporal benchmark (Section S7), the CPU–GPU computational diagnostics (Section S8) and the reproducibility and data-access arrangements (Section S9). Every number is taken from the archived aggregate results distributed with the public repository; the tables were generated from those files by the script `build_tables.py` included with this document.

S2 Source data

Table S1 lists the 32 fields of the plant workbook, the name used in the analytical table, the role each field plays in the study and whether its value can be assumed to exist at the prediction origin. Only ten fields enter the baseline: dosed mass, dosing-batch count, pelleting line, product subfamily, presentation and bag/bulk category, plus the origin hour and weekday derived from the earliest recorded stage start. Energy, duration, steam, temperature and current summaries of the current record are not used, because the workbook does not establish when they become available. The timestamps of milling and mixing define the analytical origin together with the pelleting start, whereas the completion timestamps are used only to decide which historical outcomes are admissible for the temporal analyses.

The analytical table was reconciled cell by cell with the workbook: 32 fields × 2,745 records = 87,840 comparisons, all within the documented numerical and whitespace tolerances (last column of Table S1). The three records with zero pelleting energy were kept in the audit and excluded from every supervised target. Missing values occur only in auxiliary operational summaries and in two stage timestamps; the fifteen auxiliary values that had been imputed during preparation belong to fields that are not model inputs.

Table S4 summarizes the temporal structure of the records per line and Table S5 the chronological partitions. Development and calibration eligibility require a positive recorded outcome, a valid duration, complete required inputs and completion before the next partition boundary; this removes 14 development and 11 calibration candidates, whereas all 798 test candidates are retained.

Table S1: The 32 source fields of the plant workbook, their analytical names, role in the study, availability at the prediction origin, number of missing values in the source and cells matched in the fidelity reconciliation.

Source field	Analytical name	Role	Availability at origin	Missing	Matched
DES_PRODUCTO_TIPO	salida_programada	model input	not verified before origin	0	2,745/2,745
N_BACHES_DOS	baches_dosificacion	model input	not verified before origin	0	2,745/2,745
PESO DOS (kg)	masa_dosificada_kg	model input	not verified before origin	0	2,745/2,745
PELLT	linea	model input	not verified before origin	0	2,745/2,745
CANTIDAD_BACHES_PEL	baches_peletizado	op. summary (unused)	after origin / undocumented	0	2,745/2,745
CANT PELETIZADA	produccion_t_asumida	op. summary (unused)	after origin / undocumented	0	2,745/2,745
KWH PELET	energia_peletizado_kWh	target	after origin	0	2,745/2,745
PRESION VAPOR (psi)	presion_vapor_psig	op. summary (unused)	after origin / undocumented	0	2,745/2,745
VAPOR (Lb)	masa_vapor_lb	op. summary (unused)	after origin / undocumented	0	2,745/2,745
TEMP_ACONDICIONADOR	temperatura_acond_C	op. summary (unused)	after origin / undocumented	7	2,745/2,745
AMPERAJE_PEL	corriente_peletizado_A	op. summary (unused)	after origin / undocumented	0	2,745/2,745
FEC_INI_PEL_TS	inicio_peletizado	chronology: origin	event observation	2	2,745/2,745
FEC_FIN_PEL_TS	fin_peletizado	chronology: labels	after origin	2	2,745/2,745
N_BACHES_MOL	baches_molienda	op. summary (unused)	after origin / undocumented	2	2,745/2,745
FEC_INI_MOL	inicio_molienda	chronology: origin	event observation	2	2,745/2,745
FEC_FIN_MOL	fin_molienda	chronology: labels	after origin	2	2,745/2,745
KWH MOL	energia_molienda_kWh	op. summary (unused)	after origin / undocumented	0	2,745/2,745
TEMP_DEVANADO	temperatura_devanado_C	op. summary (unused)	after origin / undocumented	5	2,745/2,745
TEMP_ROD_DELANTERO	temperatura_rod_del_C	op. summary (unused)	after origin / undocumented	5	2,745/2,745
TEMP_ROD_TRASERO	temperatura_rod_tras_C	op. summary (unused)	after origin / undocumented	5	2,745/2,745
AMPERAJE_MOL	corriente_molienda_A	op. summary (unused)	after origin / undocumented	0	2,745/2,745
N_BACHES_MEZ	baches_mezclado	op. summary (unused)	after origin / undocumented	2	2,745/2,745
FEC_INI_MEZ	inicio_mezclado	chronology: origin	event observation	2	2,745/2,745
FEC_FIN_MEZ	fin_mezclado	chronology: labels	after origin	2	2,745/2,745
KWH MEZ	energia_mezclado_kWh	op. summary (unused)	after origin / undocumented	0	2,745/2,745
AMPERAJE_MEZ	corriente_mezclado_A	op. summary (unused)	after origin / undocumented	0	2,745/2,745
UNIDADINVENTARIO	unidad_inventario	context (unused)	not verified before origin	0	2,745/2,745
AGRUPACION_INVENTARIO	grupo_inventario	context (unused)	not verified before origin	0	2,745/2,745
LINEA	familia	context (unused)	not verified before origin	0	2,745/2,745
SUBLINEA	subfamilia	model input	not verified before origin	0	2,745/2,745
PRESENTACION	forma	model input	not verified before origin	0	2,745/2,745
EMPAQUE	empaque_final	context (unused)	not verified before origin	0	2,745/2,745

Original Spanish field names are retained for traceability. Model inputs are admitted for the retrospective baseline; prospective use would require a plan version and a timestamp before the origin. The fidelity reconciliation compared every cell of the analytical table with the workbook under the documented tolerances: 32 fields × 2,745 records = 87,840 comparisons, all matched.

Table S2: Descriptive statistics of the numerical variables over all 2,745 source records (including the three zero-energy records). Pearson correlation between dosed mass and dosing-batch count: 0.979.

Variable	*n*	Mean	SD	Min	Q1	Median	Q3	Max
Dosed mass (t)	2,745	13.92	5.73	2.69	10.79	14.98	18.03	36.78
Dosing batches	2,745	3.95	1.54	1.00	3.00	5.00	5.00	10.00
Pelleting electricity (kWh)	2,745	258.16	124.07	0.00	179.10	251.10	320.40	1075.50

Table S3: Record counts of every categorical level in the 2,745 source records. Family (`LINEA`) is context only; line, subfamily, presentation and bag/bulk are model inputs.

Field	Level	Records
Pelleting line	`Pellet 2`	1,148
Pelleting line	`Pellet 3`	867
Pelleting line	`Pellet 1`	730
Product family	`LINEA PORCICULTURA`	1,242
Product family	`LINEA AVICULTURA`	1,123
Product family	`LINEA GANADERIA`	299
Product family	`LINEA EQUINOS`	55
Product family	`LINEA OTROS`	26
Product subfamily	`SUBLINEA AVICULTURA ENGORDE`	797
Product subfamily	`SUBLINEA PORCICULTURA CEBA`	783
Product subfamily	`SUBLINEA PORCICULTURA CRIA`	359
Product subfamily	`SUBLINEA AVICULTURA POSTURA`	326
Product subfamily	`SUBLINEA GANADERIA LECHE`	238
Product subfamily	`SUBLINEA PORCICULTURA PREINICIADORES`	100
Product subfamily	`SUBLINEA GANADERIA CARNE`	61
Product subfamily	`SUBLINEA EQUINOS`	55
Product subfamily	`SUBLINEA OTROS CONEJOS`	26
Presentation	`PELETIZADO`	2,065
Presentation	`CROMBELIZADO`	680
Bag/bulk	`ENSAQUE`	2,420
Bag/bulk	`GRANEL`	325

Table S4: Temporal structure of the records by pelleting line: records, positive targets, within-line overlaps with the previous interval, operations longer than 24 h, median minutes between consecutive starts and median recorded pelleting duration.

Line	Records	Positive targets	Overlaps	>24 h	Median gap (min)	Median duration (min)
Pellet 1	730	729	75	3	152.1	145.1
Pellet 2	1,148	1,146	131	6	93.0	96.0
Pellet 3	867	867	145	8	124.1	123.1

Table S5: Chronological partitions: candidate records, records retained after the eligibility rules, exclusions and the range of analytical origins (earliest recorded stage start).

Partition	Candidates	Retained	Excluded	First origin	Last origin
Development	1,556	1,542	14	2025-01-02 10:12	2025-02-21 22:34
Calibration	391	380	11	2025-02-22 00:06	2025-03-05 23:12
Test	798	798	0	2025-03-06 00:00	2025-04-01 17:28

S3 Encoding and the frozen Ridge model

Each numerical coordinate x is standardized with the training mean $\bar{x}_{\text{train}}$ and standard deviation s_{train} of the corresponding training block, $z = (x - \bar{x}_{\text{train}})/s_{\text{train}}$, and each categorical field is expanded into one indicator per level present in the training vocabulary. For the frozen model fitted on the 1,542 eligible development records this yields $6 + 3 + 9 + 2 + 2 = 22$ design columns: six numerical coordinates, three pelleting lines, nine product subfamilies, two presentations and two bag/bulk levels. A level absent from the training vocabulary activates no indicator of its field; no such level occurred in the calibration or test periods.

The model minimizes the penalized squared error of the main text with an unpenalized intercept and penalty 10 on the coefficient vector (scikit-learn `Ridge(alpha=10)` on the transformed design). Table S6 gives all 22 coefficients and Table S7 converts the numerical coefficients back to original units, $\beta_{\text{unit}} = \beta_{\text{SD}}/s_{\text{train}}$: 73.9003 kWh per training standard deviation of dosed mass (5.5556 t) is 13.3018 kWh per tonne, and 14.2309 kWh per standard deviation of batch count (1.4943 batches) is 9.5233 kWh per batch. The intercept, 288.09 kWh, is the value of the untruncated predictor at standardized means with all indicators at zero and is not a physical standby consumption. The grouped decomposition $f_i = \widehat{b} + \sum_g C_{ig}$ was recomputed for the 798 archived test estimates with a maximum absolute discrepancy of 1.14×10^{-13} kWh; no estimate required clipping at zero.

Table S6: All 22 coefficients of the frozen Ridge model (penalty 10, fitted on 1,542 development records). Intercept: 288.09 kWh. Numerical coefficients are per training standard deviation; one-hot coefficients are per indicator.

Encoded variable	Type	Coefficient (kWh)
Dosed mass	numerical (per training SD)	73.9003
Dosing batches	numerical (per training SD)	14.2309
Origin hour, sine	numerical (per training SD)	0.1912
Origin hour, cosine	numerical (per training SD)	−0.0496
Weekday, sine	numerical (per training SD)	−1.8094
Weekday, cosine	numerical (per training SD)	−1.4544
`Pellet 1`	pelleting line (one-hot)	−0.7987
`Pellet 2`	pelleting line (one-hot)	0.1181
`Pellet 3`	pelleting line (one-hot)	0.6806
`SUBLINEA AVICULTURA ENGORDE`	product subfamily (one-hot)	−10.9350
`SUBLINEA AVICULTURA POSTURA`	product subfamily (one-hot)	−27.4928
`SUBLINEA EQUINOS`	product subfamily (one-hot)	5.2334
`SUBLINEA GANADERIA CARNE`	product subfamily (one-hot)	−20.7153
`SUBLINEA GANADERIA LECHE`	product subfamily (one-hot)	11.0388
`SUBLINEA OTROS CONEJOS`	product subfamily (one-hot)	4.9565
`SUBLINEA PORCICULTURA CEBA`	product subfamily (one-hot)	−52.8982
`SUBLINEA PORCICULTURA CRIA`	product subfamily (one-hot)	23.0575
`SUBLINEA PORCICULTURA PREINICIADORES`	product subfamily (one-hot)	67.7552
`CROMBELIZADO`	presentation (one-hot)	54.3247
`PELETIZADO`	presentation (one-hot)	−54.3247
`programada_ENSAQUE`	bag/bulk (one-hot)	15.5600
`programada_GRANEL`	bag/bulk (one-hot)	−15.5600

With an unpenalized intercept and complete one-hot sets, the penalized solution centres each categorical set, which is why the two-level fields (presentation, bag/bulk) carry opposite coefficients of equal magnitude.

Table S7: Conversion of the numerical Ridge coefficients from standardized to original units: $\beta_{\text{unit}} = \beta_{\text{SD}}/s_{\text{train}}$, with the training mean and standard deviation used for standardization.

Variable	Coef. per training SD (kWh)	Training mean	Training SD	Coef. per original unit (kWh)
Dosed mass (t)	73.9003	14.1169	5.5556	13.3018
Dosing batches	14.2309	4.0208	1.4943	9.5233
Origin hour, sine	0.1912	−0.0114	0.7092	0.2696
Origin hour, cosine	−0.0496	0.0391	0.7038	−0.0704
Weekday, sine	−1.8094	0.1019	0.6747	−2.6816
Weekday, cosine	−1.4544	−0.0913	0.7252	−2.0053

S4 Reference estimators and development comparison

Three operational references were evaluated alongside the fitted models. The line-specific SEC reference multiplies the dosed mass by the ratio of total training energy to total training mass of the same line. The recent-SEC reference uses the median of the last five completed same-line energy-to-mass ratios and is the only reference whose information is updated as outcomes become available during a validation week. The line-specific median reference predicts the training median energy of the line irrespective of mass. Table S8 reports the development comparison on the 800 common targets at full archived precision, together with the three post-hoc extensions of the Ridge model: H1 adds a mass×line interaction, H2 a mass×subfamily interaction and H3 a shrunk correction from up to 20 recent same-line residuals of the frozen model ($\delta_i = \frac{n_i}{n_i+10}\bar{e}_i$ for $n_i \geq 5$, otherwise zero). Table S9 gives their paired contrasts and Figure S1 the corresponding figure.

Table S8: Development performance on the same 800 chronological targets at full archived precision: the seven original methods and the three post-hoc extensions of the Ridge model. MAE, RMSE and bias in kWh per record.

Method	n	MAE	RMSE	Bias	WAPE (%)
Ridge	800	51.4810	72.3097	−3.1361	19.88
Gradient boosting	800	51.4969	73.4650	−5.7223	19.88
ExtraTrees	800	52.3138	75.3733	−4.8856	20.20
MLP, three-seed mean	800	56.6418	81.3564	−1.4663	21.87
Line-specific SEC	800	59.6911	83.9218	−6.5940	23.05
Recent SEC, last five records	800	73.5101	109.5456	4.2886	28.38
Line-specific median	800	82.6053	112.7830	−14.0866	31.89
H1: mass by pelleting line	800	51.2644	72.1061	−3.1829	19.79
H2: mass by product subfamily	800	50.0392	71.2040	−3.8264	19.32
H3: recent-residual correction	800	49.7687	70.6723	−0.6798	19.21

Table S9: Paired MAE reductions of the exploratory extensions relative to the frozen Ridge specification on the 800 development targets: block-bootstrap percentile intervals (5,000 replicates, blocks of three observed origin dates) at the marginal 95% level and at the Bonferroni-adjusted 98.33% level for the three comparisons.

Extension	Reduction (kWh)	Reduction (%)	95% interval (kWh)	98.33% interval (kWh)
H1: mass by pelleting line	0.217	0.42	[−0.034, 0.500]	[−0.082, 0.581]
H2: mass by product subfamily	1.442	2.80	[0.701, 2.296]	[0.552, 2.481]
H3: recent-residual correction	1.712	3.33	[0.486, 2.913]	[0.236, 3.156]

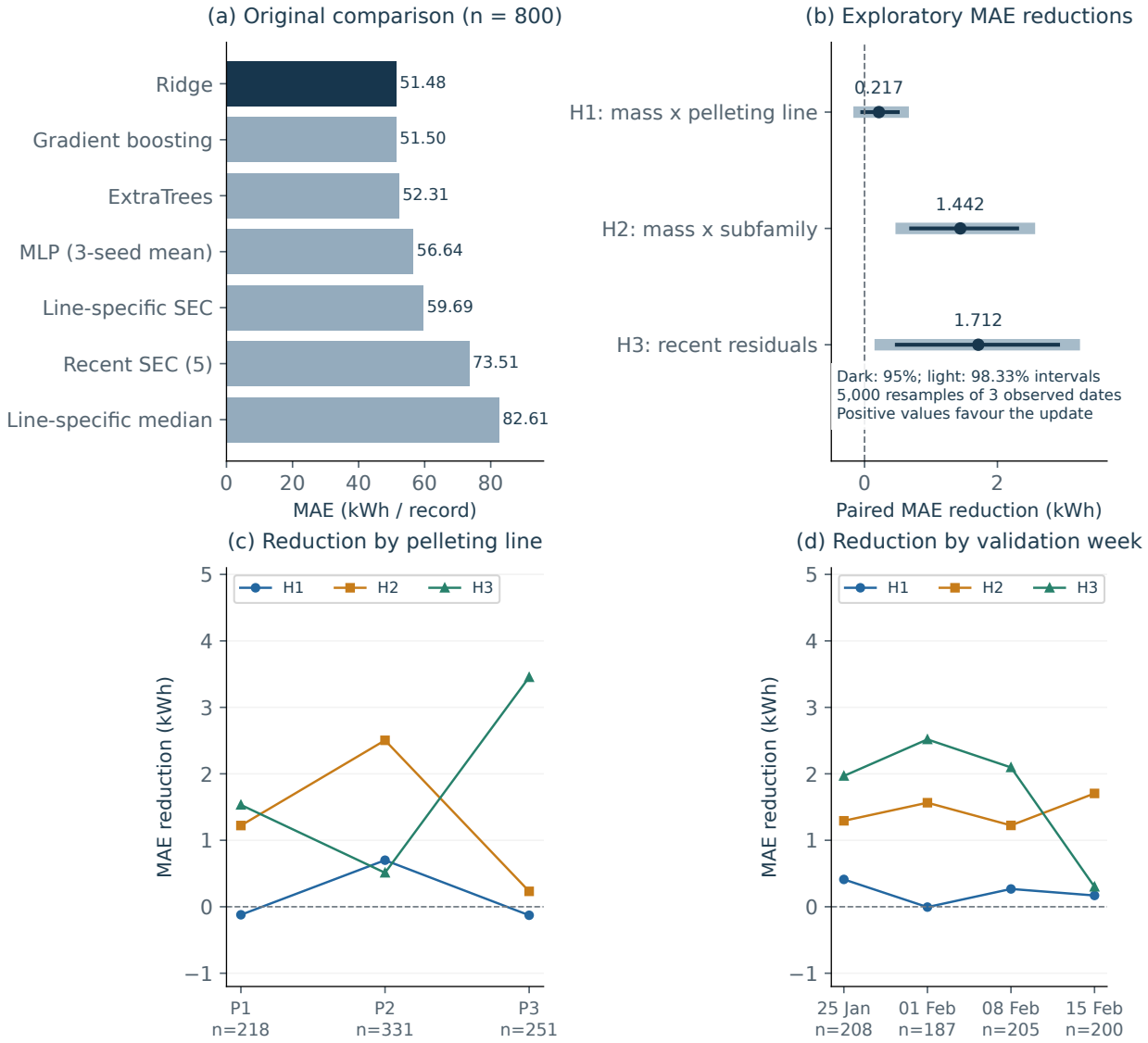


Figure S1: Development results and exploratory extensions. (a) MAE of the seven original methods on the 800 common development records. (b) Paired MAE reductions relative to Ridge for line-dependent mass effects (H1), product-dependent mass effects (H2) and recent-residual correction (H3), with 95% and multiplicity-adjusted 98.33% intervals. (c,d) Reductions by pelleting line and by validation week; positive values favour the extension.

S5 Block bootstrap and multiplicity

All paired comparisons use a circular block bootstrap over observed origin dates. Let n_d be the number of dates with at least one target (27 in the test period, 28 in the development weeks, 21 in the matched temporal benchmark). Each replicate draws $\lceil n_d/3 \rceil$ random starting dates, takes each start and the two following dates modulo n_d, truncates the sequence to n_d dates and keeps every record of every selected date, so that all lines recorded on the same date remain together. The statistic (difference of mean absolute errors, or coverage) is recomputed from the resampled records, and the percentile interval is read from the replicate distribution. The number of replicates is 3,000 for the primary test comparison (random seed 20260920) and 5,000 for the exploratory extensions (seed 20260921) and the temporal benchmark (seed 20260922). On the test period this gives a paired MAE reduction of Ridge over line-specific SEC of 8.838 kWh with 95% interval [6.076, 11.503] kWh (14.42%), and 95% intervals of [81.86, 90.24]% and [90.53, 95.87]% for the empirical coverage of the fixed 90% and 95% intervals.

Families of contrasts use approximate Bonferroni adjustment of the percentile level: $1 - 0.05/3 = 0.9833$ for the three extensions and $1 - 0.05/7 = 0.9929$ for the seven temporal contrasts. The adjustment covers the contrasts explicitly stated within each family and not the earlier exploratory choices that motivated them.

S6 Residual intervals

The frozen model's absolute calibration errors $r_j = |y_j - \widehat{y}_j|$ on the 380 calibration records are sorted, and the half-width at miscoverage α is the order statistic of rank $\lceil (n_{\text{cal}} + 1)(1 - \alpha) \rceil$: rank $\lceil 381 \times 0.90 \rceil = 343$ gives $q_{0.10} = 93.0132$ kWh and rank $\lceil 381 \times 0.95 \rceil = 362$ gives $q_{0.05} = 132.9576$ kWh. The interval of record i is $[\max(0, \widehat{y}_i - q_\alpha),\ \widehat{y}_i + q_\alpha]$.

For the worked example of the main text (source row 2080, Pellet 2, 18.002 t in five dosing batches, origin 10 March 2025 08:31), the archived estimate is 255.8964 kWh, the 90% interval [162.8833, 348.9096] kWh and the 95% interval [122.9389, 388.8540] kWh. The recorded energy, 267.3 kWh, gives a signed excess of 11.4036 kWh, both flags are zero, and the interval score at the 95% level equals the width, 265.915 kWh, because no penalty term applies. Over the 798 test records the fixed 95% intervals missed 54 outcomes (45 above the upper limit, 9 below the lower limit), i.e. 6.77 flagged records per 100 operations; the fixed 90% intervals missed 112 (88 above, 24 below).

The rolling policy keeps the point estimates fixed and recomputes the half-widths, with the same rank rule, from the 200 most recent available absolute residuals (rank 181 of 200 at the 90% level and 191 at the 95% level), ordered by recorded pelleting completion and then by source row; the pool starts with the calibration residuals and admits a test residual only after the completion of its operation precedes the origin of the target being evaluated (159,600 donor relationships checked). Tables S10 and S11 give coverage, width, interval score and exceedance counts by line and by week for both policies, and Figure S2 shows the trade-off.

Table S10: Residual intervals by pelleting line on the 798 test records: empirical coverage, mean width, interval score and the number of records above (high) and below (low) the interval, for the fixed calibration and the post-hoc rolling update.

Policy	Nominal	Line	n	Coverage (%)	Width (kWh)	Score (kWh)	High	Low
Fixed	90%	Pellet 1	200	81.00	182.96	526.08	37	1
Fixed	90%	Pellet 2	333	91.59	182.48	279.18	22	6
Fixed	90%	Pellet 3	265	82.64	185.59	401.84	29	17
Fixed	95%	Pellet 1	200	90.00	256.09	719.79	20	0
Fixed	95%	Pellet 2	333	96.40	256.38	359.20	9	3
Fixed	95%	Pellet 3	265	91.70	264.03	503.93	16	6
Rolling, 200 residuals	90%	Pellet 1	200	86.00	215.78	493.40	27	1
Rolling, 200 residuals	90%	Pellet 2	333	94.29	218.61	290.20	13	6
Rolling, 200 residuals	90%	Pellet 3	265	86.04	224.15	399.77	26	11
Rolling, 200 residuals	95%	Pellet 1	200	92.00	298.05	667.34	16	0
Rolling, 200 residuals	95%	Pellet 2	333	97.60	306.45	388.00	5	3
Rolling, 200 residuals	95%	Pellet 3	265	92.45	316.37	503.84	14	6

Table S11: Residual intervals by test week (week starting on the Monday indicated; the first and last weeks are partial).

Policy	Nominal	Week of	n	Coverage (%)	Width (kWh)	Score (kWh)	High	Low
Fixed	90%	2025-03-03	113	89.38	182.80	299.66	10	2
Fixed	90%	2025-03-10	234	91.03	183.78	243.61	15	6
Fixed	90%	2025-03-17	202	82.18	184.08	558.82	27	9
Fixed	90%	2025-03-24	206	81.07	183.00	430.92	32	7
Fixed	90%	2025-03-31	43	90.70	185.95	282.61	4	0
Fixed	95%	2025-03-03	113	92.92	255.66	363.72	6	2
Fixed	95%	2025-03-10	234	97.01	259.81	291.19	5	2
Fixed	95%	2025-03-17	202	88.61	259.16	775.02	19	4
Fixed	95%	2025-03-24	206	93.20	258.38	563.26	13	1
Fixed	95%	2025-03-31	43	95.35	262.82	355.55	2	0
Rolling, 200 residuals	90%	2025-03-03	113	92.04	192.04	301.52	7	2
Rolling, 200 residuals	90%	2025-03-10	234	88.46	168.68	250.05	19	8
Rolling, 200 residuals	90%	2025-03-17	202	85.64	244.76	535.92	23	6
Rolling, 200 residuals	90%	2025-03-24	206	91.75	263.21	424.78	15	2
Rolling, 200 residuals	90%	2025-03-31	43	95.35	244.63	300.29	2	0
Rolling, 200 residuals	95%	2025-03-03	113	93.81	270.77	364.05	6	1
Rolling, 200 residuals	95%	2025-03-10	234	94.02	224.09	309.85	9	5
Rolling, 200 residuals	95%	2025-03-17	202	92.08	350.19	734.33	13	3
Rolling, 200 residuals	95%	2025-03-24	206	97.09	378.68	579.05	6	0
Rolling, 200 residuals	95%	2025-03-31	43	97.67	318.92	347.19	1	0

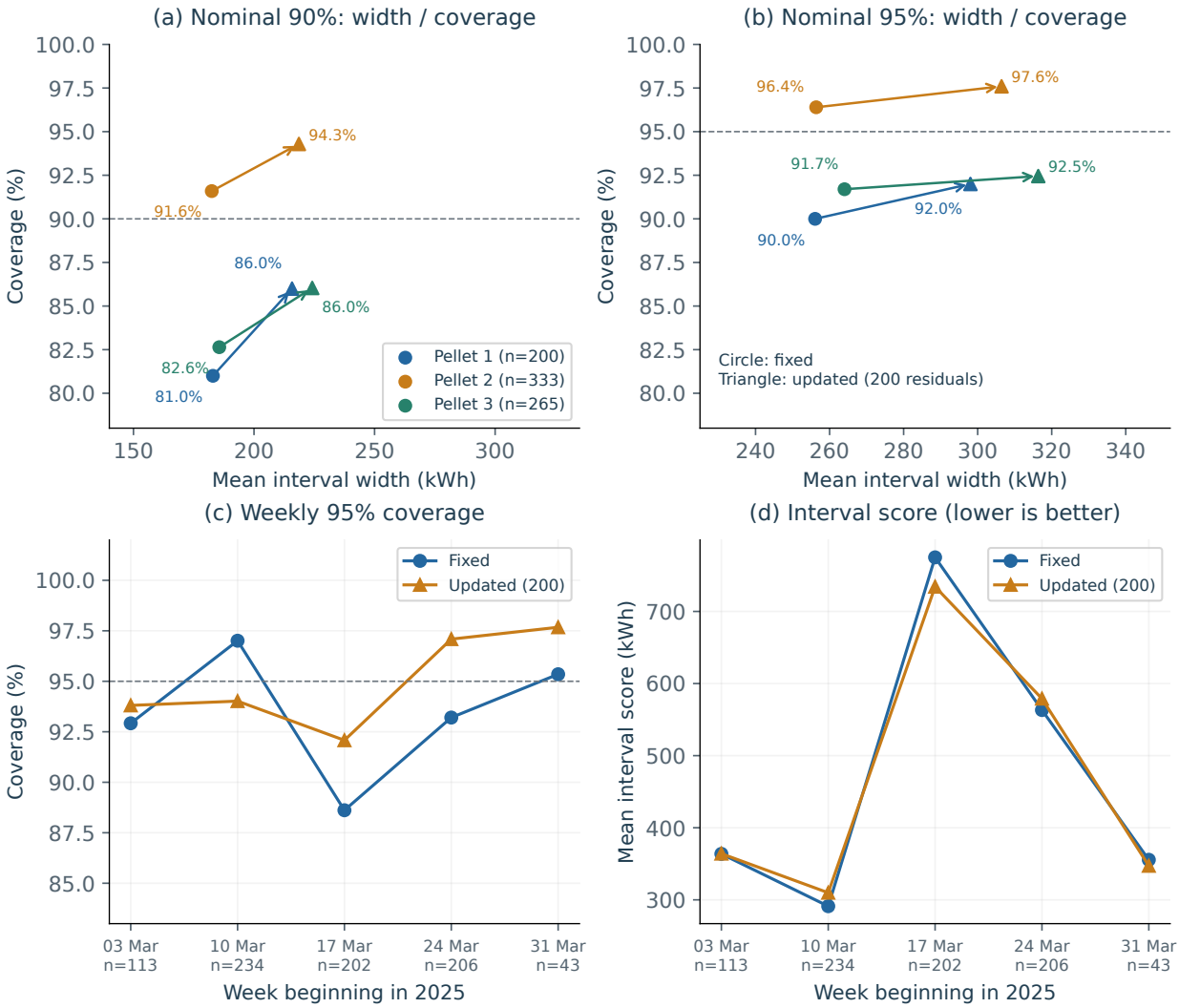


Figure S2: Coverage, width and interval score for the fixed calibration and the exploratory rolling update. (a,b) Line-specific coverage against mean interval width at nominal 90% and 95%; arrows connect fixed intervals (circles) to updated intervals (triangles). (c,d) Weekly 95% coverage and interval score, with sample sizes and partial weeks at the edges; dashed lines mark nominal coverage, and a lower interval score is better. Both policies use the same 798 fixed point predictions.

S7 Matched temporal benchmark

The benchmark asks whether the 16 most recent completed same-line operations add information beyond the current-record covariates. Contexts are built within each outer validation week and its training block, a donor must have completed before the target origin (completion is a proxy for data publication), and ties are broken by source row. Requiring 16 donors retains 584 of the 800 development targets. Table S12 lists the outer and inner sample sizes per week: the inner validation week immediately precedes the outer week and is used only to select the number of epochs of each seed (maximum 40, patience 8); each seed is then refitted from scratch on the complete outer training block and the reported prediction is the mean of the three clipped seed predictions. TFT uses hidden size 32, four attention heads and dropout 0.1; N-HiTS uses three stacks of one block each with two 64-unit layers, pooling kernels 2, 2 and 1 and no frequency downsampling. Both optimize the standardized-target MSE with Adam (learning rate 10^{-3}, weight decay 10^{-4}, batch size 64, gradient clipping at 1). The Ridge controls use penalty 10 with current covariates only or with the flattened 16-record window appended. Foundation models run zero-shot with frozen weights: TimesFM 2.5 (`google/timesfm-2.5-200m-pytorch`, revision `1d952420`) alone and with its XReg adapter (ridge 10, one call per origin so that no query pools labels from another query's future), and Chronos-2 (`amazon/chronos-2`, revision `29ec3766`) with energy only and with its native past/future covariates, cross-query learning disabled, median forecast.

Table S12: Sample sizes of the matched temporal benchmark for each outer validation week (week starting on the date shown): outer training records, inner training and inner validation records used for epoch selection, and outer validation targets.

Validation week	Outer training	Inner training	Inner validation	Targets
2025-01-25	650	447	141	151
2025-02-01	869	650	151	133
2025-02-08	1066	869	133	151
2025-02-15	1279	1066	151	149

The three extreme TimesFM XReg predictions of the first validation week (about 2,753, 9,239 and 12,203 kWh for observed values of 185.4, 142.4 and 70.9 kWh on Pellet 3) arise from contexts in which the dosed mass of the 16 donors was almost constant, so the standardized current mass lay roughly 1,100–1,467 historical standard deviations from the local mean; an independent float64 reconstruction of the adapter reproduced the extrapolation. They account for the first-week MAE of 235.9 kWh and RMSE of 1,256.3 kWh of that variant in Table S13 and are retained in every reported metric. Figures S3 and S4 show the stratified MAE and all 4,672 predictions.

Table S13: Matched temporal benchmark by outer validation week (584 targets in total). MAE, RMSE and bias in kWh per record; the last column counts raw predictions below zero before clipping.

Method	Week	n	MAE	RMSE	Bias	WAPE (%)	Neg.
Ridge, current inputs	2025-01-25	151	52.30	76.40	−11.79	19.54	0
Ridge, current inputs	2025-02-01	133	63.80	90.73	−7.07	23.01	0
Ridge, current inputs	2025-02-08	151	49.89	65.99	−9.12	19.04	0
Ridge, current inputs	2025-02-15	149	49.36	74.64	8.99	20.05	0
Ridge, full 16-record window	2025-01-25	151	67.20	85.32	−8.06	25.11	0
Ridge, full 16-record window	2025-02-01	133	74.56	99.17	3.89	26.89	1
Ridge, full 16-record window	2025-02-08	151	57.26	72.77	−8.64	21.85	1
Ridge, full 16-record window	2025-02-15	149	57.96	86.21	7.89	23.55	0
TFT	2025-01-25	151	52.28	79.77	−23.90	19.53	0
TFT	2025-02-01	133	62.73	93.12	−8.50	22.62	0
TFT	2025-02-08	151	50.59	67.02	−15.51	19.31	0
TFT	2025-02-15	149	50.04	76.06	3.52	20.33	0
N-HiTS	2025-01-25	151	75.99	104.45	−22.10	28.39	1
N-HiTS	2025-02-01	133	73.23	105.89	7.23	26.41	0
N-HiTS	2025-02-08	151	55.86	71.81	−0.14	21.32	0
N-HiTS	2025-02-15	149	56.88	85.37	4.13	23.11	0
Chronos-2 + covariates	2025-01-25	151	93.58	132.60	−9.89	34.97	0
Chronos-2 + covariates	2025-02-01	133	95.82	134.05	−17.07	34.55	0
Chronos-2 + covariates	2025-02-08	151	76.22	107.60	−5.14	29.09	0
Chronos-2 + covariates	2025-02-15	149	77.61	107.81	−1.32	31.53	0
TimesFM 2.5, energy only	2025-01-25	151	106.47	143.48	−4.31	39.78	0
TimesFM 2.5, energy only	2025-02-01	133	99.91	139.87	−17.75	36.03	0
TimesFM 2.5, energy only	2025-02-08	151	86.53	117.84	−13.66	33.02	0
TimesFM 2.5, energy only	2025-02-15	149	93.05	120.75	−2.09	37.80	0
Chronos-2, energy only	2025-01-25	151	110.41	149.51	−11.69	41.25	0
Chronos-2, energy only	2025-02-01	133	106.11	146.84	−14.29	38.27	0
Chronos-2, energy only	2025-02-08	151	91.41	124.92	−14.48	34.88	0
Chronos-2, energy only	2025-02-15	149	92.79	123.63	−6.20	37.70	0
TimesFM 2.5 + XReg	2025-01-25	151	235.92	1256.28	143.41	88.15	0
TimesFM 2.5 + XReg	2025-02-01	133	81.20	119.73	−26.73	29.28	0
TimesFM 2.5 + XReg	2025-02-08	151	59.70	83.81	−10.57	22.78	0
TimesFM 2.5 + XReg	2025-02-15	149	67.39	96.30	−0.91	27.38	0

Table S14: Matched temporal benchmark by pelleting line (149, 257 and 178 targets on Pellet 1, 2 and 3).

Method	Line	n	MAE	RMSE	Bias	WAPE (%)	Neg.
Ridge, current inputs	Pellet 1	149	48.96	78.25	−1.80	22.11	0
Ridge, current inputs	Pellet 2	257	48.13	65.50	−17.04	19.29	0
Ridge, current inputs	Pellet 3	178	65.20	90.22	10.62	20.57	0
Ridge, full 16-record window	Pellet 1	149	58.89	81.23	−4.82	26.60	0
Ridge, full 16-record window	Pellet 2	257	57.71	74.63	−13.50	23.13	2
Ridge, full 16-record window	Pellet 3	178	77.19	103.31	18.87	24.35	0
TFT	Pellet 1	149	49.21	79.60	−6.38	22.22	0
TFT	Pellet 2	257	47.22	66.22	−19.56	18.93	0
TFT	Pellet 3	178	66.65	94.32	−3.26	21.03	0
N-HiTS	Pellet 1	149	62.07	92.71	−10.29	28.03	0
N-HiTS	Pellet 2	257	56.85	76.30	−11.49	22.78	1
N-HiTS	Pellet 3	178	80.14	111.84	15.20	25.28	0
Chronos-2 + covariates	Pellet 1	149	81.78	115.21	−11.52	36.93	0
Chronos-2 + covariates	Pellet 2	257	70.10	98.97	−8.20	28.09	0
Chronos-2 + covariates	Pellet 3	178	110.94	150.47	−5.12	35.00	0
TimesFM 2.5, energy only	Pellet 1	149	97.50	132.42	−13.96	44.04	0
TimesFM 2.5, energy only	Pellet 2	257	78.83	107.38	−4.76	31.59	0
TimesFM 2.5, energy only	Pellet 3	178	120.84	157.29	−11.70	38.12	0
Chronos-2, energy only	Pellet 1	149	99.07	134.29	−16.12	44.75	0
Chronos-2, energy only	Pellet 2	257	81.34	111.66	−8.05	32.60	0
Chronos-2, energy only	Pellet 3	178	127.79	167.34	−12.96	40.31	0
TimesFM 2.5 + XReg	Pellet 1	149	65.78	101.22	−13.49	29.71	0
TimesFM 2.5 + XReg	Pellet 2	257	61.86	87.28	−9.45	24.79	0
TimesFM 2.5 + XReg	Pellet 3	178	223.48	1159.18	116.90	70.50	0

Table S15: Individual seeds of the locally trained temporal models on the 584 matched targets. The reported ensemble averages the three clipped predictions, so its MAE is not the mean of the three seed MAEs.

Method	Seed	n	MAE	RMSE	Bias	WAPE (%)
N-HiTS	11	584	71.91	97.92	−19.03	27.35
N-HiTS	29	584	70.76	98.09	11.49	26.91
N-HiTS	47	584	69.39	97.15	−1.61	26.39
TFT	11	584	55.31	79.72	−12.47	21.04
TFT	29	584	58.17	84.25	−7.78	22.13
TFT	47	584	56.43	82.42	−13.44	21.46

Table S16: Paired MAE differences relative to current-input Ridge on the 584 matched targets (positive values would favour the alternative): marginal 95% and family-adjusted 99.29% block-bootstrap intervals for the seven contrasts (5,000 replicates, blocks of three observed origin dates, seed 20260922), and the 95% interval of each method's own MAE.

Method	Improvement (kWh)	95% interval	99.29% interval	MAE 95% interval
Ridge, current inputs	0.000	[0.000, 0.000]	[0.000, 0.000]	[48.77, 59.01]
Ridge, full 16-record window	−10.403	[−13.435, −7.635]	[−14.518, −6.804]	[57.37, 70.88]
TFT	−0.105	[−2.199, 2.318]	[−2.914, 3.325]	[49.50, 58.55]
N-HiTS	−11.733	[−19.077, −6.789]	[−21.922, −5.809]	[57.76, 74.67]
Chronos-2 + covariates	−31.981	[−38.087, −27.675]	[−41.104, −26.922]	[79.03, 93.31]
TimesFM 2.5, energy only	−42.852	[−50.175, −36.525]	[−53.336, −35.112]	[90.10, 104.11]
Chronos-2, energy only	−46.477	[−54.303, −40.813]	[−57.634, −39.515]	[93.14, 108.92]
TimesFM 2.5 + XReg	−58.576	[−152.154, −14.414]	[−203.361, −11.970]	[67.58, 205.10]

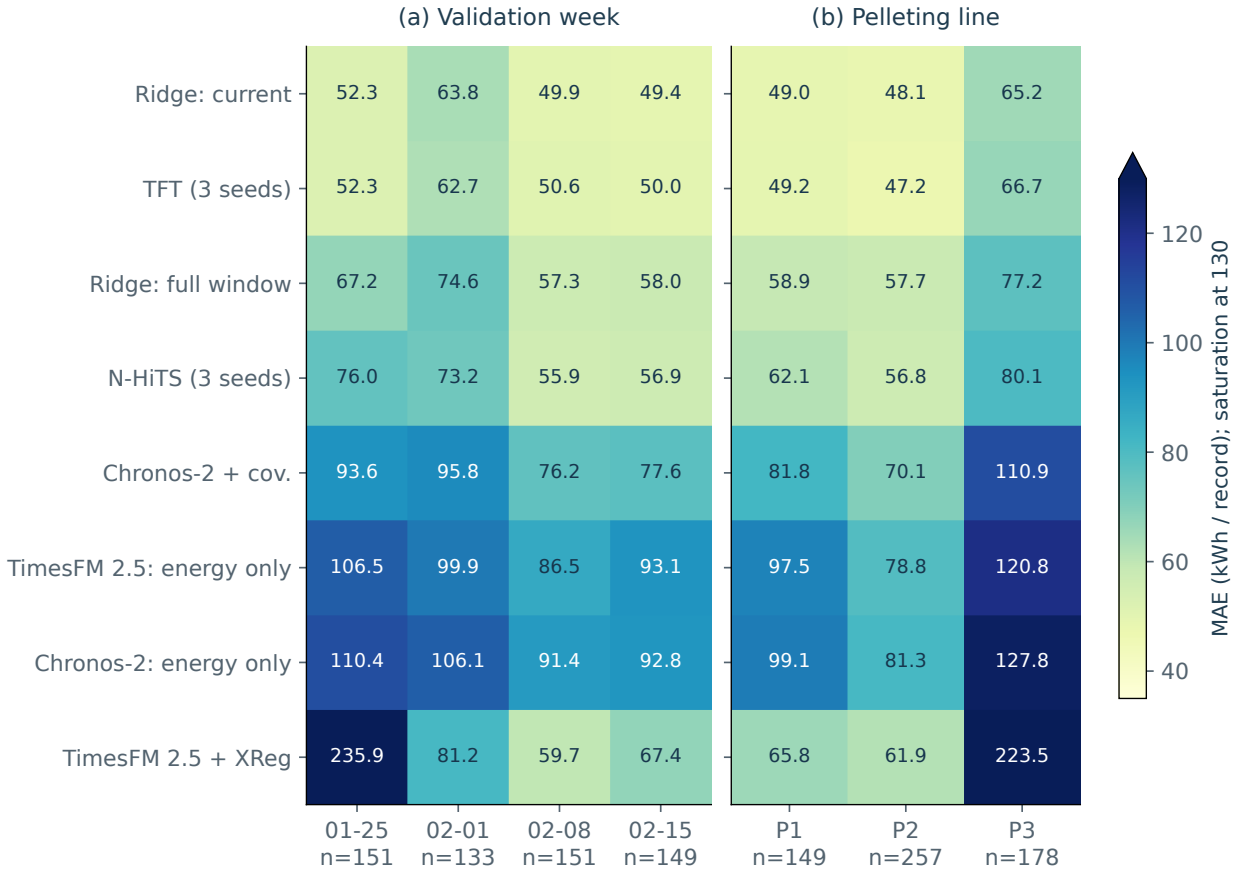


Figure S3: MAE by pelleting line and by validation week for the eight methods of the matched benchmark, in kWh per record; labels give the number of targets in each stratum. The colour scale ends at 130 kWh so that differences among the lower errors remain readable, while printed values preserve every result above that limit.

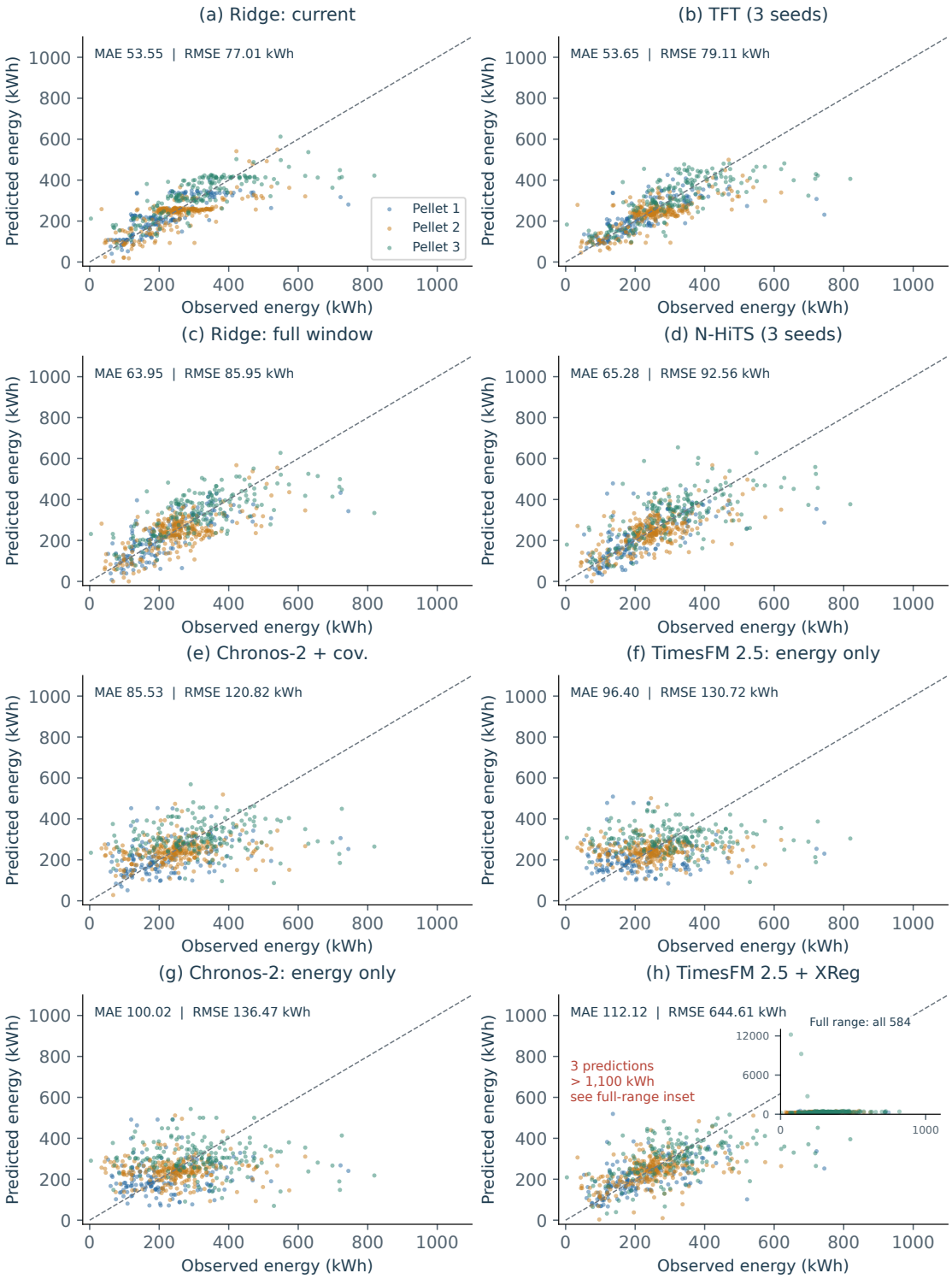


Figure S4: All 4,672 predictions of the matched comparison: eight methods on 584 identical records. Colours identify the three lines and the dashed diagonal marks equality. The main axes share a 0–1,100 kWh range; the TimesFM XReg panel adds a full-range inset containing its three extreme predictions. MAE and RMSE use the complete data in every panel.

S8 Computational diagnostics

The GPU run repeated the frozen protocol on an NVIDIA RTX PRO 5000 Blackwell (48 GB) in float32 with TF32 disabled: 24 supervised refits (two architectures × four weeks × three seeds) in 532 s of wall time and 16 foundation-model fold variants in 282 s, with a peak of 915 MiB of CUDA memory; Ridge and the JAX XReg solver ran on CPU. Table S17 compares the CPU and GPU runs, Table S18 the epochs selected on each device and Table S19 the per-method prediction differences. Within the GPU run, the 24 saved checkpoints were reloaded and their inference at batch size 8 compared with the stored batch-64 inference, together with four Chronos covariate batch-size checks: all 28 checks passed with a maximum difference of 7.63×10^{-5} kWh. Fresh CPU and GPU fits of the same seed can follow different optimization trajectories, which explains the TFT and N-HiTS differences; the frozen TimesFM variants and univariate Chronos agree across devices to below 2×10^{-4} kWh, whereas Chronos-2 with covariates differs by up to 57.04 kWh on individual records (median 0.86 kWh).

To trace the latter, the installed Chronos preprocessing and instance normalization were applied on both devices to the 584 contexts, giving 6,424 target/covariate channels. Among the 1,097 channels whose history is constant in float32, 694 showed normalized-context differences above 0.1, with a maximum of 0.8814 in the transformed (inverse-hyperbolic-sine) coordinates; no non-constant channel exceeded the threshold. The implementation substitutes a small scale only when the computed scale is exactly zero, so round-off variances of constant channels can be amplified differently by the two devices' reductions. This is a mechanism consistent with the observed prediction differences, not an end-to-end attribution of every residual, and the primary predictions were not modified.

Table S17: CPU and GPU runs of the same protocol on the 584 matched targets. TFT and N-HiTS were trained afresh on each device; foundation-model weights were frozen on both.

Method	CPU MAE	GPU MAE	CPU RMSE	GPU RMSE	GPU − CPU MAE
Ridge, current inputs	53.5457	53.5457	77.0140	77.0140	0.0000
Ridge, full 16-record window	63.9491	63.9491	85.9487	85.9487	0.0000
TFT	53.2484	53.6507	79.0792	79.1050	0.4023
N-HiTS	66.3527	65.2784	93.6651	92.5581	−1.0744
Chronos-2 + covariates	85.6480	85.5268	121.1050	120.8235	−0.1211
TimesFM 2.5, energy only	96.3980	96.3980	130.7237	130.7237	−0.0000
Chronos-2, energy only	100.0225	100.0225	136.4695	136.4695	−0.0000
TimesFM 2.5 + XReg	112.1213	112.1213	644.6068	644.6068	−0.0000

Table S18: Epoch selected on the inner validation week for every model, outer week and seed, on CPU and on GPU (maximum 40 epochs, patience 8).

Model	Week	Seed	CPU epoch	GPU epoch
TFT	2025-01-25	11	3	3
TFT	2025-01-25	29	24	14
TFT	2025-01-25	47	9	9
N-HiTS	2025-01-25	11	11	11
N-HiTS	2025-01-25	29	9	9
N-HiTS	2025-01-25	47	10	10
TFT	2025-02-01	11	2	2
TFT	2025-02-01	29	4	5
TFT	2025-02-01	47	2	2
N-HiTS	2025-02-01	11	12	12
N-HiTS	2025-02-01	29	9	9
N-HiTS	2025-02-01	47	22	30
TFT	2025-02-08	11	7	1
TFT	2025-02-08	29	6	8
TFT	2025-02-08	47	3	3
N-HiTS	2025-02-08	11	15	15
N-HiTS	2025-02-08	29	8	8
N-HiTS	2025-02-08	47	19	19
TFT	2025-02-15	11	7	7
TFT	2025-02-15	29	2	6
TFT	2025-02-15	47	5	14
N-HiTS	2025-02-15	11	11	6
N-HiTS	2025-02-15	29	9	9
N-HiTS	2025-02-15	47	13	13

Table S19: Absolute CPU–GPU prediction differences per method over the 584 matched targets (kWh).

Method	Mean	Maximum	Median
Ridge, current inputs	0.000000	0.000000	0.000000
Ridge, full 16-record window	0.000000	0.000000	0.000000
TFT	9.431759	56.837992	8.269961
N-HiTS	9.881184	47.780848	6.692332
Chronos-2 + covariates	2.386194	57.041779	0.859741
TimesFM 2.5, energy only	0.000014	0.000092	0.000000
Chronos-2, energy only	0.000015	0.000183	0.000000
TimesFM 2.5 + XReg	0.000007	0.000061	0.000000

Table S20: Software environment of the archived runs and pretrained checkpoints.

Component	Version / revision
Python	3.13.13
PyTorch	2.14.0+cu130 (CUDA 13.0)
NeuralForecast	3.2.2
timesfm	3.0.2
chronos-forecasting	2.3.2
NumPy / pandas / scikit-learn	2.5.1 / 2.3.3 / 1.9.0
TimesFM 2.5 checkpoint	`google/timesfm-2.5-200m-pytorch`
revision	`1d952420fba87f3c6dee4f240de0f1a0fbc790e3`
Chronos-2 checkpoint	`amazon/chronos-2`
revision	`29ec3766d36d6f73f0696f85560a422f50e8498c`

S9 Reproducibility and data access

The public repository https://github.com/fmarrabal/pelleting-energy-baseline-real-data (release v1.0) contains the original scientific scripts, the figure scripts, the aggregate result tables used to build this document, the frozen model configuration and the LaTeX sources of the manuscript and of this supplement. Its `reproduce.py` entry point verifies the file inventory against SHA-256 manifests, recompiles the manuscript and, once the record-level tables are in place, recomputes the metrics and regenerates every figure. Frozen artefacts are identified by fingerprint: the cleaned analytical table (`c462f92a...`), the frozen calibrated model (`b355bf28...`), the evaluation protocol (`919a53ec...`) and the source workbook (`d5968030...`).

The industrial production and electricity records, the cleaned analytical table and every record-level table derived from them (per-record predictions, residuals, intervals and contexts) belong to the plant and are not distributed publicly. They are available from the corresponding author (fmarrabal@ual.es) on reasonable request, subject to the data owner's agreement; the repository's manifests allow a recipient to verify that the received files are the ones used in the study.